

Knowledge Engineerin

A Logical Approach to Artificial Intelligence

IDENTIFICATION

CODE : IF-4-ALIA
ECTS : 2.0

HOURS

Lectures : 9.0 h
Seminars : 4.0 h
Laboratory : 12.0 h
Project : 0.0 h
Teacher-student
contact : 25.0 h
Personal work : 25.0 h
Total : 50.0 h

ASSESSMENT METHOD

A 1.5h exam will test your knowledge [documents allowed].

The project on logic programming with PROLOG will be evaluated by groups of six students: you will provide the PROLOG code you wrote and will defend your project to an audience [teachers and other students]: you will present, compare and discuss the different AI that you wrote [experimental validation].

TEACHING AIDS

All documents are available on MOODLE [<http://moodle.insa-lyon.fr>].

TEACHING LANGUAGE

French

CONTACT

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AIMS

This class aims at:

- Learn theroretical foundations of logic for Articial Intelligence: Logic of propositions, logic of predicates (first order and superior orders), revisable logics, mulit-valued logics, modal logic.
- Learn resolution and inference techniques
- Be able to model/translate problems written in natural language into logical formulas
- Be able to solve these problems with classical or non standards reasoning using inference rules
- Be able to use a logical programming language like PROLOG for modeling and solving Artificial Intelligence problems

The programming project consists in developing several Artificial Intelligence for 2-player games [reversi, chess, ...]

CONTENT

- State of the logic.
- Logic of propositions.
- Logic of predicates [first order].
- Logic of predicates [superior orders].
- Revisable logics / non-monotonous.
- Multi-valued Logics.
- Modal Logics, Alethic logic, temporal logic, epistemic logic.
- Introduction to Logic Programming.
- Introduction to PROLOG language.

BIBLIOGRAPHY

LOGIC

- [1] Nilsson N. Principes d'Intelligence Artificielle. Cepadues, Toulouse
- [2] Laurière J.C. Intelligence Artificielle, II, Représentation des Connaissances. Eyrolles
- [3] Dubois & Prade. Théorie des Possibilités. Masson, 1985
- [4] Pabion. Logique. Paris : Hermann; 1976
- [5] Thayse et coll. Approche logique de l'Intelligence Artificielle. Dunod Informatique, 1990
- [6] Tong-Tong J.R. La logique floue. Hermès, 1995

PROLOG

- [7] Blackburn P., Bos J., Striegnitz K. Learn PROLOG now! Volume 1, 2007
- [8] Nilsson U. and Maluszynski J. Logic, programming and PROLOG, 1995
- [9] <http://www.learnprolognow.org>

PRE-REQUISITE

IF-3-AAIA, IF-3-MI